

Data Analysis Midterm Project

Project Report

Field	Detail
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Date	19 July 2026
Dataset	Online Retail II (UCI Machine Learning Repository, dataset 502)
Business domain	E-commerce / Retail
Tools used	Microsoft Excel (Power Query, PivotTables) and Power BI

1. Dataset Overview

The dataset used for this project is Online Retail II, obtained from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/502/online+retail+ii>). It contains transactional sales records from a UK-based online retailer that sells mainly gift and homeware items to both individual and wholesale customers, covering the period from December 2009 to December 2011. The data is organized as two yearly files (2009–2010 and 2010–2011), which were cleaned and prepared separately and combined for analysis, giving a working dataset of 1,050,856 transaction rows across 18 columns after cleaning and feature engineering.

The business domain is e-commerce / retail. The objective of this project is to analyze the retailer's transaction history in order to answer a specific business question: which countries, products, and time periods drive the most revenue, and how much of that revenue is affected by order cancellations and returns. Revenue (calculated as Quantity × Price) was used as the primary measurable business metric throughout the analysis, tracked through the KPIs Total Orders, Total Revenue, and Average Order Value.

This dataset was selected because it is a large, genuinely messy, real-world transactional dataset with clear numerical and categorical variables, meaningful relationships between them (for example, Country and Revenue, or InvoiceDate and Revenue), and authentic data quality issues that provide a realistic opportunity to demonstrate data cleaning, exploratory analysis, and dashboard design skills using Excel and Power BI.

2. Data Cleaning

The raw dataset contained several real-world data quality issues, which were identified and addressed using Power Query in Excel. Rather than deleting problematic records outright, most issues were handled by flagging them in dedicated columns, so that the underlying data — and the option to include or exclude it from a given analysis — was preserved and documented.

Issue identified	Where	Action taken
Missing Customer ID	238,488 rows (≈22.7% of the combined dataset)	Flagged using a CustomerType column (Identified / Guest) rather than removed, since guest checkouts are a genuine, sizeable segment of this business
Cancelled and returned orders	19,432 rows combined across both years	Categorized within a Transaction Type column rather than deleted, so cancellation impact could be measured rather than hidden
Non-sale transaction types	Free Gift/Sample, Inventory Loss/Damaged, and Accounting Adjustment rows	Separated into their own Transaction Type categories so they would not distort revenue and order figures
Duplicate records	Checked across all key columns	Removed after verification
Outliers in Quantity and Price	A small number of extreme values	Flagged using an IsOutlier column instead of being deleted, keeping the row count transparent
Inconsistent text formatting	Country and Description fields	Standardized formatting (trimmed spacing, consistent casing)
Incorrect data type	InvoiceDate	Explicitly parsed as a Date/Time field rather than left as text

Beyond cleaning, several features were engineered to support the analysis and dashboard: Revenue (Quantity × Price), InvoiceYear, InvoiceMonth, DayOfWeek, and IsUK (a flag identifying United Kingdom transactions, which dominate the dataset).

Two limitations remain. First, the dataset is retained as two separate yearly tables (2009–2010 and 2010–2011) rather than a single merged table, because the 2009–2010 year is used as a historical baseline for one of the dashboard's KPI targets and keeping the years distinguishable made that comparison straightforward. Second, because roughly a quarter of transactions have no Customer ID, customer-level analysis (such as the repeat-buyer split in Section 3) necessarily under-represents guest checkouts, since repeat behavior can only be measured for identified customers.

3. Exploratory Data Analysis

The exploratory analysis focused on answering concrete business questions about the drivers of revenue, rather than producing charts for their own sake. The following patterns were found:

3.1 Seasonality

Revenue is strongly seasonal. Monthly revenue holds in a relatively narrow band between roughly 1.3M and 1.5M for most of the year, then climbs sharply from September onward and peaks in November at close to 3.0M before easing slightly in December. This Q4 (November–December) spike is by far the strongest pattern in the dataset and has direct implications for inventory and marketing planning.

3.2 Geographic concentration

The United Kingdom accounts for approximately 84% of total revenue, with the next-largest markets — Eire, the Netherlands, Germany, and France — each contributing a small fraction by comparison. Revenue is therefore highly concentrated in a single market, which represents both a strength (an established core customer base) and a risk (limited geographic diversification).

3.3 Day-of-week pattern

Revenue is not evenly distributed across the working week. Thursday generates the highest revenue of any day, ahead of Tuesday, Monday, Wednesday, and Friday, while the weekend (Saturday and Sunday) sees markedly lower activity, consistent with the business's largely wholesale/business customer base.

3.4 Customer loyalty

Among identified customers, 75.56% have placed more than one order (repeat customers), while 24.44% have ordered only once. Repeat customers therefore make up the clear majority of the identifiable customer base, indicating strong underlying customer retention once a customer is acquired and identifiable.

3.5 Product performance

Revenue is spread across a wide product catalogue, with items such as the Regency Cakestand 3 Tier, White Hanging Heart T-Light Holder, and various party and gift products (bunting, lights, ceramics) appearing among the top 10 products by revenue, reflecting the retailer's core gift and homeware positioning.

3.6 Cancellations

Cancelled or returned transactions accounted for 19,432 rows out of 1,050,856 combined transaction rows (approximately 1.85%), a relevant but modest share of overall activity, which was factored into the KPI analysis in Section 5.

4. Dashboard Summary

The interactive Power BI dashboard was built to communicate the findings above and to let a business user explore them directly, using two pages.

Page	Purpose	Visualizations
Page 1 — Performance Overview	Shows whether the business is meeting its revenue, order, and value targets, and where and when revenue is generated	KPI cards for Total Orders, Total Revenue, and Average Order Value (each against its target) · a monthly revenue trend line highlighting the Q4 spike · a country-level revenue bar chart
Page 2 — Customer & Product Detail	Shows who is buying, when, and what	A ranked customer table (Total Orders and Total Revenue per customer) · a repeat-vs-one-time customer pie chart · a revenue-by-day-of-week bar chart · a top-10-products-by-revenue bar chart

Both pages share a synchronized slicer panel (InvoiceMonth, InvoiceDate/Year, and Country), so a user can filter the entire dashboard consistently — for example, isolating a single country or comparing 2009–2010 against 2010–2011 — without losing context when moving between pages. Chart titles were written to state the finding directly (e.g., "The UK drives 84% of revenue", "Thursday drives the highest revenue", "75% of our customer base are repeat buyers") rather than simply naming the field being plotted, so the dashboard communicates insight at a glance rather than requiring the user to interpret each chart from scratch.

5. KPI Explanation

Three KPIs were built into the dashboard, each compared against an explicit target.

KPI	Measures	Target	How determined / why appropriate	Result
Total Orders	Total number of completed orders placed within the period	51,820.65	+5% above the prior period's baseline volume — a standard, realistic year-over-year growth benchmark	49,353 — 4.76% below target. Not met.
Total Revenue	Total revenue from valid transactions (Quantity × Price)	19.87M	+5% above the prior period's baseline revenue, consistent with the Orders target	20.91M — 5.26% above target. Exceeded.
Average Order Value	Total Revenue ÷ Total Orders	444.95	The 2009–2010 historical baseline — the same business a year earlier, which controls for customer mix and pricing far better than an arbitrary figure	423.76 — 4.76% below the baseline. Not met.

Read together, these three KPIs tell a specific story: revenue growth exceeded its target even though both order volume and average order value fell short of theirs. This indicates that revenue was carried disproportionately by the Q4 seasonal peak identified in Section 3.1, rather than by broad-based growth in either the number of orders placed or the value of each order across the year.

6. Business Insights and Recommendations

Insight	Recommendation
Revenue is heavily concentrated in Q4 (November–December), with the rest of the year comparatively flat	Plan inventory and staffing capacity around the Q4 surge, and test off-season promotions to smooth demand across the remaining months

Insight	Recommendation
The UK accounts for ~84% of revenue; the next-largest markets contribute far less	Consider targeted marketing investment in the next-largest markets (Eire, Netherlands, Germany, France) to reduce reliance on a single market
75.56% of identified customers are repeat buyers, but 24.44% only order once	Introduce a follow-up or loyalty offer aimed specifically at first-time customers to convert more of them into repeat buyers
Thursday generates the highest revenue of the week, with weekends notably quieter	Schedule email campaigns and promotions around the Wednesday–Thursday window to align with existing buying behavior
Average Order Value sits 4.76% below the 2009–2010 baseline, even though total revenue grew	Test bundle offers or a free-shipping threshold to lift basket size back toward the historical baseline
Cancelled/returned transactions total 19,432 rows (~1.85% of activity)	Review the specific products and countries most associated with cancellations to identify and address avoidable revenue leakage

Overall, the analysis shows a retailer with a loyal, repeat-driven customer base and a strong seasonal revenue engine, but one that is heavily dependent on both a single market (the UK) and a single time window (Q4) for its performance. The clearest opportunity identified is to broaden revenue drivers — geographically, across the calendar, and toward higher per-order value — rather than relying on the same concentrated sources of growth.